

BEYOND LIU'S 0.382709 THRESHOLD FOR THE UNION-CLOSED SETS CONJECTURE

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ABSTRACT. The union-closed sets conjecture asks whether every finite union-closed family containing a nonempty set has an element contained in at least half of its members. Let c_{UC} denote the largest universal lower bound on this proportion; equivalently, the conjecture asserts $c_{UC} = 1/2$. The best previously proved lower bound is due to Liu, who obtained a strict improvement over 0.3823455. His subsequent numerical optimization led him to conjecture the stronger threshold c_L with

$$c_L > 0.382709087918735.$$

We prove $c_{UC} \geq c_L$ and then improve this to

$$c_{UC} \geq c_L + 2 \cdot 10^{-5} > 0.382729087918735.$$

The first bound is obtained by combining the independent construction with Liu's Example 5. Liu's Example 4 gives a positive gain at the unique interior equality point, and a second convex combination turns this into a quantitative improvement over Liu's conjectured threshold.

1. INTRODUCTION

A finite family $\mathcal{F}$ of sets is *union-closed* if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$. The union-closed sets conjecture, attributed to Frankl, asserts that every finite union-closed family containing a nonempty set has an element that belongs to at least half of its members; see the survey of Bruhn and Schaudt [2]. For such a family, its *maximal element frequency* is

$$\text{freq}(\mathcal{F}) := \max_{i \in \bigcup \mathcal{F}} \frac{|\{A \in \mathcal{F} : i \in A\}|}{|\mathcal{F}|}.$$

Thus $\text{freq}(\mathcal{F})$ is the largest proportion of members of $\mathcal{F}$ that contain a fixed element. Define

$$c_{UC} := \inf_{\mathcal{F}} \text{freq}(\mathcal{F}),$$

where the infimum is over all finite union-closed families containing a nonempty set. The family $\{\emptyset, \{1\}\}$ shows that $c_{UC} \leq 1/2$, so the conjecture is equivalent to $c_{UC} = 1/2$.

Recent progress on c_{UC} is based on entropy inequalities. Gilmer [6] proved the first absolute constant, $c_{UC} \geq 0.01$. The sharp one-coordinate i.i.d. entropy inequality proved independently by Chase–Lovett [3], Alweiss–Huang–Sellke [1], and Sawin [9] yields

$$c_{UC} \geq \frac{3 - \sqrt{5}}{2} > 0.381966.$$

Sawin showed that allowing dependence gives a strict improvement, and Yu [10] and Cambie [4] made this explicit, obtaining a value about 0.3823455.

Liu [7] introduced a broader class of probabilistic constructions and analytically improved 0.3823455. His subsequent numerical optimization led him to conjecture a threshold near 0.382709 for this approach; see [7, Section V]. We refer to this value as Liu's *numerically conjectured threshold*. See also Wakhare [8] for related work on binary entropy inequalities.

Throughout, $\log$ denotes the natural logarithm. The binary entropy function is

$$H(x) := -x \log x - (1 - x) \log(1 - x), \quad 0 \leq x \leq 1,$$

with $0 \log 0 = 0$.

Let

$$p(x) := x^4 - 2x^3 + 3x^2 - 1.$$

Since

$$p'(x) = 2x(2x^2 - 3x + 3) > 0 \quad (0 < x \leq 1),$$

the polynomial p is strictly increasing on $[0, 1]$. As $p(0) = -1$ and $p(1) = 1$, it has a unique zero $x_* \in (0, 1)$. Define

$$c_L := 1 - \frac{x_* H(x_*)}{H(x_*^2)}.$$

This is the value obtained from Liu's conjectured optimizer.

Our first result proves the bound corresponding to this value.

Theorem 1.1.

$$c_{\text{UC}} \geq c_{\text{L}} > 0.382709087918735.$$

Our second result gives a strict improvement.

Theorem 1.2.

$$c_{\text{UC}} \geq c_{\text{L}} + 2 \cdot 10^{-5} > 0.382729087918735.$$

We briefly describe the proof. For Theorem 1.1, we take a convex combination of the independent construction and Liu's Example 5. The weights are chosen so that, on the diagonal, the resulting one-variable inequality has the same value and first derivative on both sides at x_* . The proof then reduces to two pointwise inequalities. One is proved analytically. The other is proved analytically outside a compact set and by a computer-assisted proof on the complementary compact region. This second inequality has a unique interior equality point.

For Theorem 1.2, we add Liu's Example 4 to the first convex combination. The estimate used for Theorem 1.1 is strict away from its unique interior equality point, while Example 4 contributes a positive amount at that point. A local strict-convexity argument handles a neighborhood of the former equality point, while analytic estimates and computational results cover the complement. This perturbative step is a quantitative counterpart of the qualitative stability-and-repair argument in Della Fiore–Costa–Dalai [5, Section 5].

Roadmap. Section 2 states Liu's one-coordinate criterion and defines the three probabilistic constructions used later. Section 3 reduces Theorem 1.1 to two pointwise inequalities. Sections 4 and 5 prove these two inequalities. Section 6 adds Liu's Example 4 and proves Theorem 1.2. Appendix A collects the computational results used in the paper; the computational supplement describes the corresponding ancillary files and verification procedures.

2. LIU'S ENTROPY CRITERION AND THE THREE CONSTRUCTIONS

The goal of this section is to state Liu's criterion that turns one-coordinate entropy inequalities into bounds for union-closed families, and to define the three constructions used later.

A *coupling* of probability measures ν_1 and ν_2 on $[0, 1]$ is a probability measure ρ on $[0, 1]^2$ whose first marginal is ν_1 and whose second marginal is ν_2 . For probability measures ν_1, ν_2 , the notation $\nu_1 \otimes \nu_2$ denotes their product measure, and $\nu^{\otimes 2} := \nu \otimes \nu$. For $s \in [0, 1]$, $\text{Ber}(s)$ denotes the Bernoulli distribution on $\{0, 1\}$ with probability s at 1.

Definition. Following Liu [7, Definition 1], a *protocol* is a family

$$\Pi = (\Pi_{s,t})_{(s,t) \in [0,1]^2}$$

such that $\Pi_{s,t}$ is a coupling of $\text{Ber}(s)$ and $\text{Ber}(t)$.

In Liu's sequential construction [7, Definition 1], a protocol is applied coordinate by coordinate to generate two copies of a prescribed $\{0, 1\}^n$ -valued random vector. At a given coordinate, let $S_i, T_i \in [0, 1]$ be the conditional probabilities that the two output bits equal 1, given their respective previous output bits. They have the same marginal distribution, say μ_i ; their joint distribution is the *induced law* at that coordinate. Conditional on $S_i = s$ and $T_i = t$, the probability that the union bit equals 0 is $\Pi_{s,t}(0, 0)$, so its entropy is $H(\Pi_{s,t}(0, 0))$.

The following result is Proposition 3 of Liu [7]. Liu states it with binary entropy in bits; multiplying by $\log 2$ gives the equivalent form below for the natural-logarithm entropy H .

Proposition 2.1 (Liu). *Let $c > 0$. For $j = 1, \dots, N$, let $\Pi^{(j)}$ be a protocol and let $w_j \geq 0$ with $\sum_j w_j = 1$. For every probability measure μ on $[0, 1]$, let $\mathcal{K}_j(\mu)$ be a nonempty set of couplings of μ with itself such that every induced law produced by $\Pi^{(j)}$ with common marginal μ belongs to $\mathcal{K}_j(\mu)$.*

Suppose that there is $\gamma > 1$ such that, for every probability measure μ on $[0, 1]$ satisfying

$$\int s \mu(ds) \leq c,$$

one has

$$\sum_{j=1}^N w_j \inf_{\rho \in \mathcal{K}_j(\mu)} \iint H(\Pi_{s,t}^{(j)}(0, 0)) \rho(ds, dt) \geq \gamma \int H(s) \mu(ds).$$

Then $c \leq c_{\text{UC}}$.

Definition. A protocol is *conditionally i.i.d.* in the sense of Liu [7, Definition 2] if it has a representation

$$\Pi_{s,t}(a, b) = \int Q_{u,s}(a) Q_{u,t}(b) \nu(du), \quad a, b \in \{0, 1\},$$

where ν is a probability measure and each $Q_{u,s}$ is a Bernoulli distribution.

Thus, for such protocols, the two output bits are independent after conditioning on the auxiliary variable u . For conditionally i.i.d. protocols, Liu [7, Definition 2 and Eq. (23)] shows that every induced law with common marginal P belongs to the class

$$(1) \quad \mathcal{C}_3(P) := \left\{ \rho : \begin{array}{l} \rho \text{ is a coupling of } P \text{ with itself, and} \\ \rho \in \overline{\text{conv}}\{Q^{\otimes 2} : Q \text{ is a Borel probability measure on } [0, 1]\} \end{array} \right\},$$

where the closure is taken for weak convergence of probability measures on $[0, 1]^2$.

The next inequality converts a product-type pointwise lower bound for a law in $\mathcal{C}_3(P)$ into an estimate depending only on its marginal P .

Lemma 2.2. *Let P be a probability measure on $[0, 1]$, let $\rho \in \mathcal{C}_3(P)$, and let $\psi : [0, 1] \rightarrow \mathbb{R}$ be continuous. Then*

$$(2) \quad \iint \psi(x)\psi(y) \rho(\mathrm{d}x, \mathrm{d}y) \geq \left(\int \psi \mathrm{d}P \right)^2.$$

Proof. By (1), there are finite mixtures

$$\rho_n = \sum_i \lambda_{i,n} Q_{i,n}^{\otimes 2}$$

that converge weakly to ρ . Put

$$P_n := \sum_i \lambda_{i,n} Q_{i,n}.$$

The first marginal of ρ_n is P_n , while the first marginal of ρ is P ; hence P_n converges weakly to P .

For each product measure $Q_{i,n}^{\otimes 2}$, the two coordinates are independent, so the integral of $\psi(x)\psi(y)$ factors. Therefore

$$\begin{aligned} \iint \psi(x)\psi(y) \rho_n(\mathrm{d}x, \mathrm{d}y) &= \sum_i \lambda_{i,n} \left(\int \psi \mathrm{d}Q_{i,n} \right)^2 \\ &\geq \left(\sum_i \lambda_{i,n} \int \psi \mathrm{d}Q_{i,n} \right)^2 = \left(\int \psi \mathrm{d}P_n \right)^2, \end{aligned}$$

where the inequality is Jensen's inequality for the convex function $z \mapsto z^2$. Since ψ is continuous on the compact interval $[0, 1]$, it is bounded. Weak convergence of ρ_n and P_n therefore allows us to pass to the limit and obtain (2). $\square$

Independent construction. The independent protocol is

$$\Pi_{s,t}^{(0)} := \text{Ber}(s) \otimes \text{Ber}(t).$$

Its two generated copies are independent. Hence, when the common marginal distribution of S_i, T_i is P , their induced law is $P \otimes P$, and

$$\Pi_{s,t}^{(0)}(0, 0) = (1-s)(1-t).$$

The other two constructions are Liu's Examples 5 and 4. Both are conditionally i.i.d. [7, Examples 4 and 5]; therefore the preceding result of Liu places their induced laws with common marginal P in $\mathcal{C}_3(P)$.

Liu's Example 5. Liu's Example 5 [7, Example 5] is specified by a measurable function $f : [0, 1] \rightarrow \mathbb{R}$ satisfying

$$0 \leq f(x) \leq \min\{x, 1-x\}.$$

For fixed $s, t \in [0, 1]$, its probability of the outcome $(0, 0)$ is

$$(3) \quad \Pi_{s,t}^{(5)}(0, 0) = (1-s)(1-t) + f(1-s)f(1-t).$$

Together with the prescribed Bernoulli marginals, (3) determines Liu's protocol; see [7, Example 5]. From now on set

$$f(x) = x(1-x).$$

Let S have distribution μ , and let P denote the distribution of $1-S$. Its mean satisfies

$$(4) \quad m(P) := \int x P(\mathrm{d}x) = 1 - \int s \mu(\mathrm{d}s).$$

Thus $\int s \mu(\mathrm{d}s) \leq c$ is equivalent to $m(P) \geq 1-c$. If $x = 1-s$ and $y = 1-t$, then

$$H(\Pi_{s,t}^{(0)}(0, 0)) = H(xy).$$

For Liu's Example 5, equation (3) gives

$$H(\Pi_{s,t}^{(5)}(0, 0)) = H(r(x, y)),$$

where

$$(5) \quad r(x, y) := xy(1 + (1 - x)(1 - y)), \quad g(x) := r(x, x) = x^2(1 + (1 - x)^2).$$

Liu's Example 4. Liu's Example 4 [7, Example 4] is defined by

$$(6) \quad a(t) = \begin{cases} 0, & 0 \leq t \leq 1 - 1/\sqrt{2}, \\ \sqrt{\frac{1 - 2(1 - t)^2}{2t(1 - t)}}, & 1 - 1/\sqrt{2} < t \leq 1/2, \\ 1, & 1/2 < t \leq 1. \end{cases}$$

Its probability of the outcome $(0, 0)$ is

$$(7) \quad \Pi_{s,t}^{(4)}(0, 0) = (1 - s)(1 - t) + a(s)a(t)(\min\{1 - s, 1 - t\} - (1 - s)(1 - t)).$$

Together with the prescribed Bernoulli marginals, (7) defines Liu's protocol; see [7, Example 4]. The three branches in (6) agree at the two breakpoints, so a is continuous.

3. PROOF OF LIU'S CONJECTURED THRESHOLD

The goal of this section is to prove Theorem 1.1 by applying Liu's criterion to the two protocols $\Pi^{(0)}$ and $\Pi^{(5)}$ with suitable weights. The proof reduces to two pointwise inequalities, established in Sections 4 and 5, which are then integrated to verify the criterion.

Recall x_* and c_L from the introduction. Define

$$(8) \quad m_* := 1 - c_L = \frac{x_* H(x_*)}{H(x_*^2)}.$$

From (5),

$$g(x) = 2x^2 - 2x^3 + x^4.$$

Since $p(x_*) = 0$, we have

$$x_*^2 + g(x_*) = 1,$$

and the symmetry $H(u) = H(1 - u)$ gives

$$(9) \quad H(g(x_*)) = H(x_*^2).$$

Choose $\alpha + \beta = 1$ so that, on the diagonal $x = y$, the two sides of

$$m_* [\alpha H(x^2) + \beta H(g(x))] \geq xH(x)$$

have the same derivative at $x = x_*$. Equations (9) and (8) already give equality of their values at x_* . With

$$A_0(x) := H(x^2), \quad B_0(x) := H(g(x)),$$

this derivative condition is linear in β . Solving it gives

$$(10) \quad \beta := \frac{A_0(x_*) \left(\frac{1}{x_*} + \frac{H'(x_*)}{H(x_*)} \right) - A'_0(x_*)}{B'_0(x_*) - A'_0(x_*)}, \quad \alpha := 1 - \beta.$$

The interval bound for β in Proposition A.1 implies $\alpha, \beta \in (0, 1)$, so these are valid convex-combination weights.

By equation (5), the integrand from Liu's Example 5 becomes $H(r(x, y))$ in complemented coordinates. The first pointwise inequality lower-bounds it by the geometric mean of its diagonal values at x and y .

Lemma 3.1. *For every $x, y \in [0, 1]$,*

$$(11) \quad H(r(x, y))^2 \geq H(g(x))H(g(y)).$$

If $x, y \in (0, 1)$, equality holds if and only if $x = y$.

The second pointwise inequality combines the independent integrand $H(xy)$ with the two diagonal values $H(g(x))$ and $H(g(y))$ from Example 5.

Lemma 3.2. *For every $x, y \in [0, 1]$,*

$$(12) \quad 2\alpha H(xy) + 2\beta \sqrt{H(g(x))H(g(y))} \geq \frac{xH(y) + yH(x)}{m_*}.$$

In $(0, 1)^2$, equality holds only at (x_, x_*) . On the boundary of $[0, 1]^2$, equality holds exactly when $xy = 0$ and at $(1, 1)$.*

Lemmas 3.1 and 3.2 are proved in Sections 4 and 5. Together they give the fixed-marginal estimate required by Liu's criterion.

Put

$$(13) \quad b(x) := \sqrt{H(g(x))}.$$

The function b is continuous on $[0, 1]$.

Lemma 3.1 and Lemma 2.2 give a lower bound for the Example 5 integral that depends only on the marginal distribution.

Lemma 3.3. *Let P be a probability measure on $[0, 1]$ and let $\rho \in \mathcal{C}_3(P)$. Then*

$$(14) \quad \iint H(r(x, y)) \rho(\mathrm{d}x, \mathrm{d}y) \geq \left(\int b(x) P(\mathrm{d}x) \right)^2.$$

Proof. Since both sides of (11) are nonnegative, taking square roots gives

$$H(r(x, y)) \geq b(x)b(y).$$

Integrating this inequality against ρ and applying Lemma 2.2 with $\psi = b$ gives (14). $\square$

The next proposition combines the independent and Example 5 integrals into the fixed-marginal estimate needed in Liu's criterion.

Proposition 3.4. *Let P be a probability measure on $[0, 1]$ and let $\rho \in \mathcal{C}_3(P)$. Then*

$$(15) \quad \alpha \iint H(xy) P(\mathrm{d}x)P(\mathrm{d}y) + \beta \iint H(r(x, y)) \rho(\mathrm{d}x, \mathrm{d}y) \geq \frac{m(P)}{m_*} \int H(x) P(\mathrm{d}x).$$

Proof. Integrate (12) with respect to $P(\mathrm{d}x)P(\mathrm{d}y)$. The first term becomes

$$2\alpha \iint H(xy) P(\mathrm{d}x)P(\mathrm{d}y),$$

and the second becomes, by (13),

$$2\beta \left(\int b(x) P(\mathrm{d}x) \right) \left(\int b(y) P(\mathrm{d}y) \right) = 2\beta \left(\int b \mathrm{d}P \right)^2.$$

For the right-hand side, the integrand is bounded on $[0, 1]^2$, because $0 \leq x, y \leq 1$ and $0 \leq H \leq \log 2$. It is therefore integrable with respect to $P \otimes P$, so Fubini's theorem gives

$$\begin{aligned} \iint \frac{xH(y) + yH(x)}{m_*} P(\mathrm{d}x)P(\mathrm{d}y) &= \frac{1}{m_*} \int x \mathrm{d}P \int H(y) P(\mathrm{d}y) \\ &\quad + \frac{1}{m_*} \int y \mathrm{d}P \int H(x) P(\mathrm{d}x) \\ &= \frac{2m(P)}{m_*} \int H \mathrm{d}P. \end{aligned}$$

Dividing by 2 therefore yields

$$(16) \quad \alpha \iint H(xy) \mathrm{d}P \mathrm{d}P + \beta \left(\int b \mathrm{d}P \right)^2 \geq \frac{m(P)}{m_*} \int H \mathrm{d}P.$$

By Lemma 3.3,

$$\iint H(r(x, y)) \mathrm{d}\rho \geq \left(\int b \mathrm{d}P \right)^2.$$

Substitution in (16) proves (15). $\square$

Proof of Theorem 1.1. We verify the hypothesis of Liu's criterion, Proposition 2.1. Fix $c < c_L$ and let μ be a probability measure on $[0, 1]$ with

$$\int s \mu(\mathrm{d}s) \leq c.$$

If S has distribution μ , let P be the distribution of $1 - S$. By (4),

$$m(P) \geq 1 - c > 1 - c_L = m_*.$$

Set

$$\gamma := \frac{1 - c}{m_*} > 1.$$

We apply Proposition 2.1 with $N = 2$, protocols $\Pi^{(0)}$ and $\Pi^{(5)}$, and weights α and β . In the notation of that proposition, take

$$\mathcal{K}_0(\mu) := \{\mu \otimes \mu\}, \quad \mathcal{K}_5(\mu) := \mathcal{C}_3(\mu).$$

The weights are nonnegative and satisfy $\alpha + \beta = 1$. Since $\mathcal{K}_0(\mu) = \{\mu \otimes \mu\}$ is a singleton, its infimum in Liu's criterion equals the unique integral. Thus the required left-hand side is

$$\alpha \iint H(\Pi_{s,t}^{(0)}(0,0)) \mu(ds) \mu(dt) + \beta \inf_{\rho \in \mathcal{C}_3(\mu)} \iint H(\Pi_{s,t}^{(5)}(0,0)) \rho(ds, dt).$$

Under the reflection $x = 1 - s$, $y = 1 - t$, the first integral becomes $\iint H(xy) P(dx) P(dy)$. The same reflection sends $\mathcal{C}_3(\mu)$ bijectively to $\mathcal{C}_3(P)$, because it sends every product measure to a product measure and commutes with convex combinations and weak limits. For Example 5, the second integrand becomes $H(r(x, y))$. Also $H(1 - s) = H(s)$ gives $\int H dP = \int H d\mu$. Proposition 3.4 therefore implies that this convex combination is at least

$$\frac{m(P)}{m_*} \int H dP \geq \gamma \int H dP = \gamma \int H d\mu,$$

where the inequality uses $m(P) \geq 1 - c$ and $\int H dP \geq 0$.

Proposition 2.1 now gives $c \leq c_{UC}$. Since this holds for every $c < c_L$, taking the supremum over such c proves

$$c_{UC} \geq c_L.$$

□

4. PROOF OF THE FIRST POINTWISE INEQUALITY

The goal of this section is to prove Lemma 3.1. The proof splits according to the value of $g(y)$. When $g(y) < 1/2$, Lemmas 4.1, 4.2, and 4.3 show that a mixed derivative of $\log H(r(x, y))$ is negative. When $g(y) \geq 1/2$, Lemma 4.4 gives the needed comparison directly. We establish these four ingredients first and then prove Lemma 3.1.

For $u \in (0, 1)$, where $H(u) > 0$, set

$$\Lambda(u) := \log H(u), \quad M(u) := \frac{uH'(u)}{H(u)}.$$

The following one-variable inequality for H will be used in the case $g(y) < 1/2$.

Lemma 4.1. *For $0 < u < 1/2$,*

$$(17) \quad -\frac{\Lambda''(u)}{\Lambda'(u)} > \frac{1}{u} + \frac{1}{1-u}.$$

Proof. For $0 < u < 1/2$, put

$$\ell := \log \frac{1-u}{u} > 0, \quad q := -\log(1-u) > 0.$$

Then

$$H'(u) = \ell, \quad H(u) = u\ell + q,$$

and therefore

$$(18) \quad M(u) = \frac{u\ell}{u\ell + q}.$$

For $0 < u < 1/2$, both $M(u)$ and $\Lambda'(u) = H'(u)/H(u)$ are positive. Since $M = u\Lambda'$, we may take logarithms:

$$\log M = \log u + \log \Lambda'.$$

Differentiating term by term gives

$$\frac{M'}{M} = \frac{1}{u} + \frac{\Lambda''}{\Lambda'}.$$

Hence

$$-\frac{\Lambda''}{\Lambda'} = -\frac{M'}{M} + \frac{1}{u},$$

and therefore

$$-\frac{\Lambda''}{\Lambda'} > \frac{1}{u} + \frac{1}{1-u} \iff -\frac{M'}{M} > \frac{1}{1-u}.$$

Thus it remains to prove

$$(19) \quad -\frac{M'}{M} > \frac{1}{1-u}.$$

We compute the left-hand side explicitly. Since

$$\ell' = -\frac{1}{u(1-u)},$$

taking logarithms in (18) gives

$$\log M = \log u + \log \ell - \log H.$$

Differentiating term by term gives

$$\frac{M'}{M} = \frac{1}{u} + \frac{\ell'}{\ell} - \frac{H'}{H} = \frac{1}{u} - \frac{1}{u(1-u)\ell} - \frac{\ell}{H}.$$

Hence

$$\begin{aligned} -\frac{M'}{M} - \frac{1}{1-u} &= -\frac{1}{u} + \frac{1}{u(1-u)\ell} + \frac{\ell}{H} - \frac{1}{1-u} \\ &= \frac{1-\ell}{u(1-u)\ell} + \frac{\ell}{H} \\ &= \frac{H(1-\ell) + u(1-u)\ell^2}{u(1-u)\ell H} \\ (20) \quad &= \frac{q(1-\ell) + u\ell(1-u\ell)}{u(1-u)\ell H}. \end{aligned}$$

The denominator in (20) is positive, so it remains to show that the numerator is positive.

If $0 < \ell \leq 1$, then $q(1-\ell) \geq 0$. Moreover $0 < u\ell \leq u < 1/2$, so $u\ell(1-u\ell) > 0$. Thus the numerator is positive.

Assume now $\ell > 1$. The elementary estimate

$$-\log(1-u) \leq \frac{u}{1-u}$$

gives $q \leq u/(1-u)$. Because $1-\ell < 0$, multiplying the inequality $q \leq u/(1-u)$ by $1-\ell$ reverses the inequality sign and gives

$$q(1-\ell) \geq \frac{u}{1-u}(1-\ell).$$

Therefore

$$(21) \quad q(1-\ell) + u\ell(1-u\ell) \geq \frac{u}{1-u} (1 - u\ell(\ell+1) + u^2\ell^2).$$

Now

$$u = \frac{1}{1+e^\ell} < e^{-\ell}.$$

For $\ell \geq 1$ we have $e^\ell > \ell(\ell+1)$. To see this without any approximation, use the cubic Taylor polynomial:

$$e^\ell > 1 + \ell + \frac{\ell^2}{2} + \frac{\ell^3}{6}.$$

Subtracting $\ell(\ell+1)$ gives

$$1 - \frac{\ell^2}{2} + \frac{\ell^3}{6}.$$

Its derivative is $\ell(\ell/2 - 1)$, so its minimum on $[1, \infty)$ is attained at $\ell = 2$ and equals $1/3$. Hence it is positive. Consequently

$$u\ell(\ell+1) < 1.$$

The parenthesized factor in (21) is therefore strictly positive: the sum of its first two terms is $1 - u\ell(\ell+1) > 0$, and its last term $u^2\ell^2$ is nonnegative. This proves (19), hence (17). $\square$

From (5),

$$r(x, y) = xy(2 - x - y + xy).$$

Define

$$W := 2 - x - y + xy, \quad U := 2 - 2x - y + 2xy, \quad V := 2 - x - 2y + 2xy.$$

Thus $r = xyW$, and direct differentiation gives

$$(22) \quad r_x = yU, \quad r_y = xV, \quad r_{xy} = 2 - 2x - 2y + 4xy.$$

For $0 < x, y < 1$, one has $W = 1 + (1-x)(1-y) > 0$, $U = y + 2(1-x)(1-y) > 0$, and $V = x + 2(1-x)(1-y) > 0$. Hence r is strictly increasing in each variable on $(0, 1)^2$, and in particular $0 < r(x, y) < 1$ there.

The next identity rewrites the mixed derivative of r in the form needed to combine it with Lemma 4.1.

Lemma 4.2. *For $0 < x, y < 1$, with W, U, V defined above,*

$$(23) \quad \frac{r_{xy}}{r_x r_y} = \frac{1}{r} + \frac{1}{WUV}.$$

Proof. Substitute $r = xyW$, $r_x = yU$, and $r_y = xV$. By the positivity just established, multiplication by $xyWUV$ preserves the equality and reduces (23) to

$$W r_{xy} = UV + xy.$$

Using the definitions of W, U, V and (22),

$$\begin{aligned} W r_{xy} &= (2 - x - y + xy)(2 - 2x - 2y + 4xy), \\ UV + xy &= (2 - 2x - y + 2xy)(2 - x - 2y + 2xy) + xy. \end{aligned}$$

Both sides expand to

$$4 - 6x - 6y + 2x^2 + 2y^2 + 14xy - 6x^2y - 6xy^2 + 4x^2y^2.$$

□

Let y_0 be the unique number in $(0, 1)$ with $g(y_0) = 1/2$. Its existence follows from $g(0) = 0 < 1/2 < 1 = g(1)$ and continuity. Indeed,

$$g'(x) = 4x - 6x^2 + 4x^3 = 2x(2 - 3x + 2x^2) > 0 \quad (0 < x < 1),$$

so g is strictly increasing. The calculation

$$g(43/64) - \frac{1}{2} = \frac{305}{2^{24}} > 0$$

shows that

$$y_0 < \frac{43}{64}.$$

The following polynomial inequality allows us to compare the second term in Lemma 4.2 with the right-hand side of Lemma 4.1 on the square containing $[0, y_0]^2$.

Lemma 4.3. *For $0 \leq x, y \leq 43/64$,*

$$(24) \quad WUV \geq 1 - r(x, y).$$

Proof. After the rescaling

$$x = \frac{43}{64}s, \quad y = \frac{43}{64}t, \quad 0 \leq s, t \leq 1,$$

the polynomial $WUV - (1 - r)$ has bidegree $(3, 3)$. Proposition A.2 gives its exact coefficient matrix in the product Bernstein basis of degree $(3, 3)$; every coefficient is strictly positive. Since every Bernstein basis function is nonnegative on $[0, 1]^2$, the polynomial is nonnegative there. Rescaling back proves (24). □

For the case $g(y) \geq 1/2$, define

$$G(v) := e^v H(e^{-v}) \quad (v > 0).$$

The following monotonicity property of M handles this case.

Lemma 4.4. *The function $M(u) = uH'(u)/H(u)$ is strictly decreasing on $(0, 1)$.*

Proof. Put $u = e^{-v}$. By the definition above,

$$G(v) = \frac{H(u)}{u}, \quad H(e^{-v}) = e^{-v}G(v).$$

Hence

$$\log H(e^{-v}) = -v + \log G(v).$$

Using $-\log(1 - z) = \sum_{k \geq 1} z^k/k$ with $z = e^{-v}$ gives

$$G(v) = v + 1 - \sum_{k \geq 1} \frac{e^{-kv}}{k(k+1)}.$$

Termwise differentiation yields

$$G''(v) = -\sum_{k \geq 1} \frac{ke^{-kv}}{k+1} < 0.$$

Since $G(v) > 0$,

$$(\log G)'' = \frac{G''}{G} - \left(\frac{G'}{G}\right)^2 < 0 :$$

the first term is strictly negative and the second is nonpositive. Thus $\log G$ is strictly concave. The function $v \mapsto -v$ is affine, and adding an affine function preserves strict concavity; hence $v \mapsto \log H(e^{-v}) = -v + \log G(v)$ is strictly concave.

Its derivatives can be computed directly. Since $u = e^{-v}$ and $u' = -u$,

$$\frac{d}{dv} \log H(e^{-v}) = \frac{H'(u)}{H(u)}(-u) = -\frac{uH'(u)}{H(u)} = -M(u).$$

Differentiating once more,

$$\frac{d^2}{dv^2} \log H(e^{-v}) = -M'(u)u' = uM'(u) = e^{-v}M'(e^{-v}).$$

Strict concavity makes this quantity negative. Since $e^{-v} > 0$, we obtain $M'(e^{-v}) < 0$, and therefore $M'(u) < 0$ for every $0 < u < 1$. □

Proof of Lemma 3.1. Both sides of (11) are continuous on $[0, 1]^2$, so the boundary inequality follows once the inequality is proved in the open square. On the diagonal, $g(x) = r(x, x)$ gives equality. Thus it remains only to prove the strict inequality for $x \neq y$ in the open square. By symmetry assume $0 < x < y < 1$ and define

$$\mathcal{L}(x, y) := \log H(r(x, y))$$

and

$$(25) \quad \Delta_x(y) := \mathcal{L}(x, y) - \frac{1}{2}\mathcal{L}(x, x) - \frac{1}{2}\mathcal{L}(y, y).$$

Since $g(z) = r(z, z)$, the strict inequality in Lemma 3.1 is

$$2\mathcal{L}(x, y) > \mathcal{L}(x, x) + \mathcal{L}(y, y),$$

which is exactly $\Delta_x(y) > 0$. Moreover $\Delta_x(x) = 0$. Thus it suffices to prove $\Delta'_x(s) > 0$ for every $s \in (x, y]$, because then

$$\Delta_x(y) = \Delta_x(x) + \int_x^y \Delta'_x(s) ds > 0.$$

After renaming s as y , the remaining goal is therefore

$$\Delta'_x(y) > 0 \quad (0 < x < y < 1).$$

Differentiate (25) with respect to y . The term $\mathcal{L}(x, x)$ is constant, while symmetry gives $\mathcal{L}_1(y, y) = \mathcal{L}_2(y, y)$. Hence

$$(26) \quad \Delta'_x(y) = \mathcal{L}_2(x, y) - \mathcal{L}_2(y, y).$$

We split the proof into two cases according to the value of $g(y)$.

Case A: $g(y) < 1/2$. Then $y < y_0$, and for every $s \in [x, y]$ we have $s, y \leq y_0$ and

$$r(s, y) \leq r(y, y) = g(y) < 1/2.$$

By definition, $\mathcal{L}(x, y) = \log H(r(x, y)) = \Lambda(r(x, y))$, so $\mathcal{L} = \Lambda \circ r$. The chain rule gives

$$\mathcal{L}_{12} = \Lambda''(r)r_x r_y + \Lambda'(r)r_{xy} = \Lambda'(r)r_x r_y \left(\frac{r_{xy}}{r_x r_y} + \frac{\Lambda''(r)}{\Lambda'(r)} \right).$$

Here $\Lambda'(r) = H'(r)/H(r) > 0$ because $r < 1/2$, and $r_x, r_y > 0$. By Lemmas 4.2 and 4.3,

$$\frac{r_{xy}}{r_x r_y} = \frac{1}{r} + \frac{1}{WUV} \leq \frac{1}{r} + \frac{1}{1-r}.$$

Lemma 4.1 gives

$$\frac{\Lambda''(r)}{\Lambda'(r)} < -\frac{1}{r} - \frac{1}{1-r}.$$

Therefore $\mathcal{L}_{12}(s, y) < 0$ for every $s \in [x, y]$. Apply the fundamental theorem of calculus to the function $s \mapsto \mathcal{L}_2(s, y)$:

$$\mathcal{L}_2(y, y) - \mathcal{L}_2(x, y) = \int_x^y \mathcal{L}_{12}(s, y) ds < 0.$$

Thus $\mathcal{L}_2(x, y) > \mathcal{L}_2(y, y)$, and equation (26) gives $\Delta'_x(y) > 0$.

Case B: $g(y) \geq 1/2$. By Lemma 4.4, M is strictly decreasing on $(0, 1)$.

Next, differentiating $\mathcal{L}$ with respect to its second variable gives

$$(27) \quad \mathcal{L}_2(x, y) = \frac{r_y(x, y)}{r(x, y)} M(r(x, y)) = A_x(y) M(r(x, y)),$$

where

$$(28) \quad A_x(y) := \frac{1}{y} - \frac{1-x}{1+(1-x)(1-y)}.$$

For $0 < x < y < 1$, the derivative r_y is positive, so $A_x(y) > 0$. Moreover the function

$$u \mapsto \frac{u}{1 + u(1 - y)}$$

is strictly increasing for $u \geq 0$. Since $1 - x > 1 - y$, equation (28) gives

$$(29) \quad 0 < A_x(y) < A_y(y).$$

Also $r(x, y) < r(y, y) = g(y)$ because r is strictly increasing in its first variable.

If $r(x, y) < 1/2 \leq g(y)$, then

$$M(r(x, y)) > 0 \geq M(g(y)),$$

so (27) and (29) immediately give

$$\mathcal{L}_2(x, y) > \mathcal{L}_2(y, y).$$

If $1/2 \leq r(x, y) < g(y)$, then the strict decrease of M gives

$$M(g(y)) < M(r(x, y)) \leq 0.$$

Therefore

$$A_x(y)M(r(x, y)) > A_x(y)M(g(y)) > A_y(y)M(g(y)),$$

where the second inequality reverses because $M(g(y)) < 0$ and $A_x(y) < A_y(y)$. Again $\mathcal{L}_2(x, y) > \mathcal{L}_2(y, y)$. Thus $\Delta'_x(y) > 0$ in every case.

Both cases give $\Delta'_x(y) > 0$. Hence the integral identity following (25) gives $\Delta_x(y) > 0$, which is equivalent to the strict form of (11) by the definition of Δ_x . Equality on the diagonal was noted at the start, and continuity gives the boundary inequality. $\square$

5. PROOF OF THE SECOND POINTWISE INEQUALITY

The goal of this section is to prove Lemma 3.2. We first rewrite its inequality as $F(t, d) \geq 0$. Two analytic lemmas prove positivity in three boundary regions. We then locate the equality point and use the computational results of Appendix A on the compact part left over. These regions cover the entire domain $t > 0, 0 \leq d \leq t$.

By symmetry it is enough to consider $0 < x \leq y < 1$. Introduce logarithmic coordinates

$$(30) \quad x = e^{-(t+d)}, \quad y = e^{-(t-d)}, \quad t > 0, \quad 0 \leq d < t.$$

Recall from Section 4 that

$$G(u) = e^u H(e^{-u}) = \frac{H(e^{-u})}{e^{-u}} \quad (u > 0),$$

and define

$$\Gamma_5(u) := \frac{\sqrt{H(g(e^{-u}))}}{e^{-u}}.$$

Both functions have continuous extensions at $u = 0$, with $G(0) = \Gamma_5(0) = 0$. Define

$$(31) \quad F(t, d) := m_* [\alpha G(2t) + \beta \Gamma_5(t+d) \Gamma_5(t-d)] - \frac{G(t+d) + G(t-d)}{2}.$$

The next proposition records the reduction used throughout the section.

Proposition 5.1. *For $0 < x \leq y < 1$, the change of variables (30) is a bijection with $t > 0, 0 \leq d < t$, and inequality (12) is equivalent to $F(t, d) \geq 0$. The formula (31) extends continuously to $d = t$, so below we prove $F(t, d) \geq 0$ on the larger domain $t > 0, 0 \leq d \leq t$.*

Proof. From (30), $xy = e^{-2t}$ and

$$\frac{H(xy)}{xy} = G(2t).$$

Moreover,

$$\frac{\sqrt{H(g(x))H(g(y))}}{xy} = \frac{\sqrt{H(g(e^{-(t+d)}))}}{e^{-(t+d)}} \frac{\sqrt{H(g(e^{-(t-d)}))}}{e^{-(t-d)}} = \Gamma_5(t+d) \Gamma_5(t-d),$$

and

$$\frac{xH(y) + yH(x)}{2xy} = \frac{1}{2} \left(\frac{H(y)}{y} + \frac{H(x)}{x} \right) = \frac{G(t-d) + G(t+d)}{2}.$$

Multiplying (12) by the positive factor $m_*/(2xy)$ gives

$$m_* \left[\alpha \frac{H(xy)}{xy} + \beta \frac{\sqrt{H(g(x))H(g(y))}}{xy} \right] - \frac{1}{2} \left(\frac{H(y)}{y} + \frac{H(x)}{x} \right) \geq 0.$$

Substituting the three identities just proved turns the left-hand side into

$$m_* [\alpha G(2t) + \beta \Gamma_5(t+d) \Gamma_5(t-d)] - \frac{G(t+d) + G(t-d)}{2} = F(t, d).$$

Thus (12) is equivalent to $F(t, d) \geq 0$. $\square$

To obtain the two analytic boundary estimates, we need monotonicity, concavity, and elementary bounds for G .

Lemma 5.2. *The function G is increasing and concave on $[0, \infty)$, and strictly increasing and strictly concave on $(0, \infty)$. For every $u \geq 0$,*

$$u + 1 - e^{-u} \leq G(u) \leq u + 1.$$

Proof. Starting from the definition of H with $x = e^{-u}$,

$$G(u) = \frac{-x \log x - (1-x) \log(1-x)}{x} = u + (e^u - 1) [-\log(1 - e^{-u})].$$

Using $-\log(1-z) = \sum_{k \geq 1} z^k/k$ at $z = e^{-u}$ and multiplying by $e^u - 1$ gives

$$(32) \quad G(u) = u + 1 - \sum_{k \geq 1} \frac{e^{-ku}}{k(k+1)}.$$

The series and its first two derivatives converge locally uniformly for $u > 0$, so termwise differentiation is valid there. We obtain

$$G'(u) = 1 + \sum_{k \geq 1} \frac{e^{-ku}}{k+1} > 0, \quad G''(u) = -\sum_{k \geq 1} \frac{ke^{-ku}}{k+1} < 0.$$

This proves strict monotonicity and strict concavity on $(0, \infty)$. Together with $G(0) = 0$, these properties extend in the non-strict form to $[0, \infty)$.

Finally, $e^{-ku} \leq e^{-u}$ for $k \geq 1$, and

$$\sum_{k \geq 1} \frac{1}{k(k+1)} = \sum_{k \geq 1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1.$$

Hence

$$0 \leq \sum_{k \geq 1} \frac{e^{-ku}}{k(k+1)} \leq e^{-u} \sum_{k \geq 1} \frac{1}{k(k+1)} = e^{-u}.$$

Writing the series in (32) as $S(u)$, the estimate just obtained is $0 \leq S(u) \leq e^{-u}$. Hence

$$u + 1 - e^{-u} \leq u + 1 - S(u) = G(u) \leq u + 1,$$

which gives the two stated bounds. $\square$

We first prove strict positivity when t is small.

Lemma 5.3. *For $0 < t \leq 0.002$ and $0 \leq d \leq t$ one has $F(t, d) > 0$.*

Proof. Put

$$c_0 := m_* \alpha.$$

The interval bounds in Proposition A.1 imply

$$c_0 > 0.5555293761467 > \frac{1}{2}.$$

The two arguments $t+d$ and $t-d$ have average t . Since G is concave by Lemma 5.2,

$$\frac{G(t+d) + G(t-d)}{2} \leq G(t).$$

The factor β and the function Γ_5 are nonnegative, so the corresponding term in (31) can be discarded in a lower bound. Hence

$$(33) \quad F(t, d) \geq c_0 G(2t) - G(t).$$

We estimate the two values of G explicitly.

The elementary inequality $e^t - 1 \leq te^t$ and the lower bound $e^t - 1 \geq t$ imply

$$1 - e^{-t} = e^{-t}(e^t - 1) \geq te^{-t}.$$

Since $-\log$ is decreasing on $(0, \infty)$, from $1 - e^{-t} \geq te^{-t} > 0$ we get

$$-\log(1 - e^{-t}) \leq -\log(te^{-t}) = -\log t + t.$$

Using the formula for G at the start of Lemma 5.2,

$$(34) \quad G(t) \leq t + te^t(-\log t + t).$$

For $2t$, we use $e^{2t} - 1 \geq 2t$ and $1 - e^{-2t} \leq 2t$. The latter gives $-\log(1 - e^{-2t}) \geq -\log(2t)$, and therefore

$$(35) \quad G(2t) \geq 2t + 2t[-\log(2t)] = 2t(1 - \log(2t)).$$

Define

$$\eta(t) := 2c_0(1 - \log(2t)) - 1 - e^t(-\log t + t).$$

Substituting (34) and (35) into (33) gives

$$F(t, d) \geq t\eta(t).$$

Differentiation gives

$$\eta'(t) = \frac{e^t - 2c_0}{t} - e^t(1 - \log t + t).$$

For $0 < t \leq 0.002$, we have $e^t < 1.003 < 2c_0$. Thus the first term is negative. The second is also negative because $1 - \log t + t > 0$. Hence $\eta'(t) < 0$ throughout the interval, so η is minimized at $t = 0.002$. Proposition A.1 gives the certified lower bound

$$\eta(0.002) \geq 0.016672474402352748 > 0.$$

Since $t > 0$, the inequality $F(t, d) \geq t\eta(t)$ now proves the claim. $\square$

The next estimate covers points close to the edge $d = t$ and all sufficiently large t .

Lemma 5.4. *Let $v = t - d$. Then $F(t, d) > 0$ whenever one of the following holds:*

- (a) $0.002 \leq t \leq 1.4$ and $0 \leq v \leq 5 \cdot 10^{-5}$;
- (b) $1.4 \leq t \leq 4.1$ and $0 \leq v \leq 0.1$;
- (c) $t \geq 4.1$ and $0 \leq d \leq t$.

Proof. Set

$$u := t + d, \quad v := t - d.$$

Then $u + v = 2t$ and $0 \leq v \leq t \leq u \leq 2t$. From (31),

$$F(t, d) = c_0 G(2t) + m_* \beta \Gamma_5(u) \Gamma_5(v) - \frac{1}{2} G(u) - \frac{1}{2} G(v).$$

Since $m_*, \beta, \Gamma_5(u), \Gamma_5(v)$ are nonnegative, deleting their product gives

$$F(t, d) \geq c_0 G(2t) - \frac{1}{2} G(u) - \frac{1}{2} G(v).$$

Also $u \leq 2t$ and G is increasing, so $G(u) \leq G(2t)$. Therefore

$$F(t, d) \geq \left(c_0 - \frac{1}{2}\right) G(2t) - \frac{1}{2} G(v).$$

The last expression is increasing in t and decreasing in v , because $c_0 > 1/2$ and G is increasing. Thus its minimum on each of the two rectangles occurs at the smallest allowed t and the largest allowed v . The corresponding certified interval evaluations, recorded in Proposition A.1, are

$$\left(c_0 - \frac{1}{2}\right) G(0.004) - \frac{1}{2} G(5 \cdot 10^{-5}) \geq 0.0011788376979667077$$

and

$$\left(c_0 - \frac{1}{2}\right) G(2.8) - \frac{1}{2} G(0.1) \geq 0.035598095197518845.$$

The two certified endpoint evaluations prove parts (a) and (b).

For part (c), Lemma 5.2 gives

$$G(2t) \geq 2t + 1 - e^{-2t}.$$

It also gives $G(u) \leq u + 1$ and $G(v) \leq v + 1$, so, using $u + v = 2t$,

$$\frac{G(u) + G(v)}{2} \leq \frac{(u + 1) + (v + 1)}{2} = t + 1.$$

Consequently

$$(36) \quad F(t, d) \geq c_0(2t + 1 - e^{-2t}) - (t + 1).$$

The derivative of the lower bound in (36) is

$$2c_0 + 2c_0 e^{-2t} - 1 > 2c_0 - 1 > 0,$$

so this lower bound is increasing in t . At the left endpoint $t = 4.1$,

$$2t + 1 - e^{-2t} = 9.2 - e^{-8.2}, \quad t + 1 = 5.1.$$

Thus Proposition A.1 gives

$$c_0(9.2 - e^{-8.2}) - 5.1 \geq 0.010717682423498683 > 0.$$

Hence (36) is positive for every $t \geq 4.1$, proving part (c). $\square$

To treat the interior, we first identify the stationary point corresponding to the equality configuration $x = y = x_*$ of Lemma 3.2.

Let

$$t_* := -\log x_*.$$

Lemma 5.5. *The point $(t_*, 0)$ satisfies:*

- (a) $F(t_*, 0) = 0$;
- (b) $F_t(t_*, 0) = 0$;
- (c) $F_d(t_*, 0) = 0$.

Proof. For part (c), the function $F(t, d)$ is even in d , as is clear from (31); hence $F_d(t, 0) = 0$ and in particular $F_d(t_*, 0) = 0$.

For parts (a) and (b), put $x = e^{-t}$ and define

$$\Delta_0(x) := m_*[\alpha H(x^2) + \beta H(g(x))] - xH(x).$$

Setting $d = 0$ in (31) and using $G(t) = H(x)/x$ and $\Gamma_5(t)^2 = H(g(x))/x^2$ gives

$$(37) \quad F(t, 0) = e^{2t}\Delta_0(e^{-t}).$$

At $x = x_*$, equation (9), the identity $\alpha + \beta = 1$, and the definition of m_* give

$$\Delta_0(x_*) = m_*H(x_*^2) - x_*H(x_*) = 0.$$

Thus $F(t_*, 0) = 0$, proving part (a).

To prove part (b), differentiate (37). Writing $x = e^{-t}$, so $x' = -x$, gives

$$F_t(t, 0) = e^{2t}(2\Delta_0(x) - x\Delta_0'(x)).$$

At $t = t_*$ the first term vanishes because $\Delta_0(x_*) = 0$. Since $e^{2t_*}x_* > 0$, we have

$$F_t(t_*, 0) = 0 \iff \Delta_0'(x_*) = 0.$$

Finally,

$$\Delta_0'(x) = m_*[\alpha A_0'(x) + \beta B_0'(x)] - H(x) - xH'(x).$$

At $x = x_*$, substitute $m_* = x_*H(x_*)/A_0(x_*)$ and $\alpha = 1 - \beta$. After division by the positive factor $x_*H(x_*)/A_0(x_*)$, the equation $\Delta_0'(x_*) = 0$ becomes

$$A_0'(x_*) + \beta(B_0'(x_*) - A_0'(x_*)) = A_0(x_*) \left(\frac{1}{x_*} + \frac{H'(x_*)}{H(x_*)} \right),$$

which is exactly the defining equation (10) for β . Hence $F_t(t_*, 0) = 0$, proving part (b). $\square$

The two boundary lemmas leave a compact region. The next result combines the analytic estimates above with Proposition A.1 to cover it and to show that $(t_*, 0)$ is the only equality point.

Lemma 5.6. *For every $t > 0$ and $0 \leq d \leq t$,*

$$F(t, d) \geq 0,$$

with equality if and only if $(t, d) = (t_, 0)$.*

Proof. To apply the computational results of Proposition A.1, we first express the second derivatives of F in terms of derivatives of G and Γ_5 .

We begin with G . Put $x = e^{-u}$. Then $x' = -x$, and

$$H'(x) = \log \frac{1-x}{x}, \quad H''(x) = -\frac{1}{x(1-x)}.$$

Since $G(u) = e^u H(x)$, the product and chain rules give

$$G'(u) = e^u H(x) + e^u H'(x)(-x) = G(u) - H'(x).$$

Differentiating again,

$$G''(u) = G'(u) - H''(x)x' = G'(u) - \frac{1}{1-x}.$$

For Γ_5 , keep $x = e^{-u}$ and put $z = g(x)$. The chain rule gives

$$z' = -xg'(x), \quad z'' = xg'(x) + x^2g''(x).$$

Since $\Gamma_5(u) = e^u \sqrt{H(z)}$, another application of the product and chain rules gives

$$\Gamma_5'(u) = \Gamma_5(u) \left(1 + \frac{H'(z)z'}{2H(z)} \right).$$

For the second derivative, differentiate the parenthesized factor as well. Because

$$\frac{d}{du} \frac{H'(z)z'}{H(z)} = \frac{H''(z)(z')^2 + H'(z)z''}{H(z)} - \left(\frac{H'(z)z'}{H(z)} \right)^2,$$

we obtain

$$\begin{aligned} \Gamma_5''(u) = \Gamma_5(u) & \left[\left(1 + \frac{H'(z)z'}{2H(z)} \right)^2 \right. \\ & \left. + \frac{1}{2} \left(\frac{H''(z)(z')^2 + H'(z)z''}{H(z)} - \left(\frac{H'(z)z'}{H(z)} \right)^2 \right) \right]. \end{aligned}$$

Finally, calculate the second derivatives of F . With $u = t + d$ and $v = t - d$, differentiating (31) twice gives

$$\begin{aligned} F_{tt} &= 4m_*\alpha G''(2t) + m_*\beta(\Gamma_5''(u)\Gamma_5(v) + 2\Gamma_5'(u)\Gamma_5'(v) + \Gamma_5(u)\Gamma_5''(v)) \\ &\quad - \frac{1}{2}(G''(u) + G''(v)), \\ F_{dd} &= m_*\beta(\Gamma_5''(u)\Gamma_5(v) - 2\Gamma_5'(u)\Gamma_5'(v) + \Gamma_5(u)\Gamma_5''(v)) \\ &\quad - \frac{1}{2}(G''(u) + G''(v)), \\ F_{td} &= m_*\beta(\Gamma_5''(u)\Gamma_5(v) - \Gamma_5(u)\Gamma_5''(v)) - \frac{1}{2}(G''(u) - G''(v)). \end{aligned}$$

Define the local rectangle

$$(38) \quad \mathcal{Q}_0 := \{(t, d) : |t - 0.36992289186771087| \leq 0.06, \ 0 \leq d \leq 0.06\}.$$

Proposition A.1 gives throughout $\mathcal{Q}_0$

$$F_{tt} \geq 0.13459156779665421, \quad F_{dd} \geq 0.87181314699157286,$$

and

$$F_{tt}F_{dd} - F_{td}^2 \geq 0.050114365218749776 > 0.$$

Thus the Hessian is positive definite throughout $\mathcal{Q}_0$, and F is strictly convex there. By Lemma 5.5, $(t_*, 0)$ is stationary and has value 0. Strict convexity therefore makes it the unique minimum, and the unique zero, in $\mathcal{Q}_0$.

It remains to cover precisely the part of the triangle not handled by Lemma 5.3, Lemma 5.4, or the local rectangle. To work with rectangular cells, set

$$\xi := d/t \in [0, 1], \quad u = t(1 + \xi), \quad v = t(1 - \xi),$$

and define $\tilde{F}(t, \xi) := F(t, t\xi)$. The change $d = t\xi$ maps the triangular condition $0 \leq d \leq t$ to the fixed interval $0 \leq \xi \leq 1$.

The first derivatives used in the finite estimates follow from the ordinary chain rule. Since

$$\frac{\partial}{\partial t} F(t, t\xi) = F_t(t, t\xi) + \xi F_d(t, t\xi), \quad \frac{\partial}{\partial \xi} F(t, t\xi) = t F_d(t, t\xi),$$

substitution of (31) gives

$$\begin{aligned} \tilde{F}_t &= 2m_*\alpha G'(2t) + m_*\beta((1 + \xi)\Gamma_5'(u)\Gamma_5(v) + (1 - \xi)\Gamma_5(u)\Gamma_5'(v)) \\ &\quad - \frac{1}{2}((1 + \xi)G'(u) + (1 - \xi)G'(v)), \\ \tilde{F}_\xi &= m_*\beta t(\Gamma_5'(u)\Gamma_5(v) - \Gamma_5(u)\Gamma_5'(v)) - \frac{t}{2}(G'(u) - G'(v)). \end{aligned}$$

We call a *box* a closed axis-parallel rectangle

$$R = [t_0 - \Delta_t, t_0 + \Delta_t] \times [\xi_0 - \Delta_\xi, \xi_0 + \Delta_\xi].$$

For any $(t, \xi) \in R$, apply the one-variable fundamental theorem of calculus first along the t direction and then along the ξ direction. Taking absolute values gives

$$(39) \quad \tilde{F}(t, \xi) \geq \tilde{F}(t_0, \xi_0) - \sup_R |\tilde{F}_t| \Delta_t - \sup_R |\tilde{F}_\xi| \Delta_\xi.$$

Thus a lower bound for the center value and upper bounds for the two derivatives give a lower bound for $\tilde{F}$ on the whole box.

After the analytic boundary regions are removed, the remaining points have $v = t(1 - \xi)$ bounded below as follows:

t -range	v lower bound
$[0.002, 0.1]$	$5 \cdot 10^{-5}$
$[0.1, 0.34]$	$5 \cdot 10^{-5}$
$[0.34, 0.40]$	$5 \cdot 10^{-5}$
$[0.40, 1.4]$	$5 \cdot 10^{-5}$
$[1.4, 4.1]$	0.1

Proposition A.1 gives, for each row of the table, a finite cover by such boxes. Boxes mapping into $\mathcal{Q}_0$ are covered by the strict-convexity argument above. On every remaining box the right-hand side of (39) has a certified positive lower bound; the smallest among the five regional minima in Proposition A.1 is

$$1.2396204165999228 \cdot 10^{-6} > 0.$$

Hence $\tilde{F} > 0$ on every such box. Together with the three analytic regions of Lemmas 5.3 and 5.4, these covers exhaust the domain $t > 0, 0 \leq d \leq t$. Thus $F(t, d) \geq 0$ everywhere, with equality only at $(t_*, 0)$. $\square$

Proof of Lemma 3.2. By Proposition 5.1, it is enough to prove, in the open square, $F(t, d) \geq 0$. Lemma 5.6 proves this and shows that equality occurs only at $(t, d) = (t_*, 0)$, which corresponds to $(x, y) = (x_*, x_*)$.

It remains to inspect the boundary. If $xy = 0$, say $x = 0$, then $H(xy) = H(0) = 0$, $g(0) = 0$, and

$$xH(y) + yH(x) = 0 \cdot H(y) + yH(0) = 0;$$

the same argument applies when $y = 0$. Hence both sides of (12) are zero along these two boundary edges. At $(1, 1)$, all entropy factors are again zero.

Suppose next that $x = 1$ and $0 < y < 1$. Since $g(1) = 1$, the factor $H(g(1))$ vanishes, and (12) reduces, after cancelling the positive number $H(y)$, to

$$2\alpha \geq \frac{1}{m_*}.$$

The interval bounds in Proposition A.1 give

$$2\alpha - \frac{1}{m_*} > 0.1799 > 0,$$

so the inequality is strict. The case $y = 1, 0 < x < 1$ is symmetric. We have therefore obtained exactly the boundary equality set stated in Lemma 3.2. $\square$

6. ADDING LIU'S EXAMPLE 4

The goal of this section is to prove Theorem 1.2. We first obtain a product lower bound for the Example 4 term in Liu's criterion. We then combine Example 4 with the two constructions used in Section 3 and prove a new pointwise inequality. The last part applies Liu's criterion with these three constructions.

6.1. A lower bound for Example 4. We use complemented coordinates. Thus, for the original parameters $s, t \in [0, 1]$, put

$$x := 1 - s, \quad y := 1 - t,$$

and define

$$\tilde{a}(x) := a(1 - x).$$

For $x, y \in [0, 1]$ set

$$K_4(x, y) := H(xy + \tilde{a}(x)\tilde{a}(y)(\min\{x, y\} - xy)).$$

Equation (7) then gives

$$K_4(x, y) = H(\Pi_{1-x, 1-y}^{(4)}(0, 0)).$$

By the reflection argument in the proof of Theorem 1.1, $\mathcal{C}_3(\mu)$ is sent bijectively to $\mathcal{C}_3(P)$, where P is the distribution of $1 - S$ for $S \sim \mu$. Hence the Example 4 term in Liu's criterion can be studied through

$$\iint K_4(x, y) \rho(dx, dy), \quad \rho \in \mathcal{C}_3(P).$$

The following auxiliary function is chosen so that $K_4(x, y)$ has a product lower bound. Set

$$x_{\text{cut}} := \frac{181}{256}, \quad \sigma := \frac{99999}{100000},$$

and define the cutoff

$$\chi(x) := \begin{cases} 0, & 0 \leq x \leq 1/64, \\ \frac{x - 1/64}{3/64}, & 1/64 < x < 1/16, \\ 1, & 1/16 \leq x \leq 15/16, \\ \frac{63/64 - x}{3/64}, & 15/16 < x < 63/64, \\ 0, & 63/64 \leq x \leq 1. \end{cases}$$

The cutoff makes the auxiliary function vanish near 0 and 1. We also use

$$D(x) := \begin{cases} H(x), & 0 \leq x \leq 1/2, \\ \log 2, & 1/2 \leq x \leq x_{\text{cut}}, \\ \frac{\log 2}{H(x_{\text{cut}}^2)} H(x^2), & x_{\text{cut}} \leq x \leq 1. \end{cases}$$

The adjacent formulas agree at $1/2$ and x_{cut} , so D is continuous. Finally, define

$$\varphi(x) := \sigma \chi(x) \sqrt{\frac{D(x)H(x)}{H(x_*)}} \exp\left(-\frac{H'(x_*)}{2H(x_*)}(x - x_*) - \frac{7}{10}(x - x_*)^2\right).$$

The functions χ , D , and H are continuous and nonnegative on $[0, 1]$, so φ is continuous and nonnegative.

The next result turns the two-variable Example 4 entropy term $K_4(x, y)$ into a product of one-variable terms.

Lemma 6.1. *For all $x, y \in [0, 1]$,*

$$(40) \quad K_4(x, y) \geq \varphi(x)\varphi(y).$$

Proposition A.3 in Appendix A proves Lemma 6.1 computationally.

The following consequence of Lemmas 6.1 and 2.2 is the Example 4 estimate that will be used in the proof of Theorem 1.2.

Proposition 6.2. *Let P be a probability measure on $[0, 1]$ and let $\rho \in \mathcal{C}_3(P)$. Then*

$$\iint K_4(x, y) \rho(\mathrm{d}x, \mathrm{d}y) \geq \left(\int \varphi(x) P(\mathrm{d}x) \right)^2.$$

Proof. By Lemma 6.1,

$$\iint K_4(x, y) \rho(\mathrm{d}x, \mathrm{d}y) \geq \iint \varphi(x)\varphi(y) \rho(\mathrm{d}x, \mathrm{d}y).$$

Apply Lemma 2.2 with $\psi = \varphi$ to bound the last integral by $(\int \varphi \mathrm{d}P)^2$. □

6.2. The three-construction pointwise inequality. Set

$$(41) \quad \theta := \frac{111}{2500}, \quad \varepsilon_{\text{cert}} := \frac{41}{2000000}, \quad m_{\text{cert}} := m_* - \varepsilon_{\text{cert}}.$$

Thus $\theta = 0.0444$ and $\varepsilon_{\text{cert}} = 2.05 \cdot 10^{-5}$.

The following pointwise inequality will be used in Liu's criterion.

Lemma 6.3. *For all $x, y \in [0, 1]$,*

$$(42) \quad \begin{aligned} (1 - \theta) \left(2\alpha H(xy) + 2\beta \sqrt{H(g(x))H(g(y))} \right) + 2\theta \varphi(x)\varphi(y) \\ \geq \frac{xH(y) + yH(x)}{m_{\text{cert}}}. \end{aligned}$$

Proof. For $0 < x \leq y < 1$, use the logarithmic coordinates of (30), so

$$x = e^{-(t+d)}, \quad y = e^{-(t-d)}, \quad t > 0, \quad 0 \leq d < t,$$

and define

$$\Phi(u) := e^u \varphi(e^{-u}).$$

For the fixed parameters in (41), set

$$(43) \quad \mathcal{N}(t, d) := m_{\text{cert}} \left[(1 - \theta) (\alpha G(2t) + \beta \Gamma_5(t + d) \Gamma_5(t - d)) \right. \\ \left. + \theta \Phi(t + d) \Phi(t - d) \right] - \frac{G(t + d) + G(t - d)}{2}.$$

Multiplying (42) by the positive factor $m_{\text{cert}}/(2xy)$ shows that it is equivalent to

$$\mathcal{N}(t, d) \geq 0.$$

Indeed, the independent, Example 5, and Example 4 terms become respectively

$$m_{\text{cert}}(1 - \theta)\alpha G(2t), \quad m_{\text{cert}}(1 - \theta)\beta \Gamma_5(t + d) \Gamma_5(t - d), \quad m_{\text{cert}}\theta \Phi(t + d) \Phi(t - d),$$

while the right-hand side becomes $[G(t + d) + G(t - d)]/2$.

Write $d = t\xi$, where $0 \leq \xi \leq 1$, and put

$$u := t(1 + \xi), \quad v := t(1 - \xi),$$

and

$$F_\varepsilon(t, \xi) := \mathcal{N}(t, t\xi), \quad c_\varepsilon := m_{\text{cert}}(1 - \theta)\alpha.$$

It is therefore enough to show

$$F_\varepsilon(t, \xi) \geq 0 \quad (t > 0, 0 \leq \xi \leq 1).$$

We use the following five regions. The first three are handled analytically; the last two use Propositions A.5 and A.6.

method	region
analytic	$0 < t \leq 10^{-5}$,
analytic	$t \geq 8$,
analytic	$\left(c_\varepsilon - \frac{1}{2}\right) G(2t) - \frac{1}{2} G(v) > 0$,
computer-assisted	$\mathcal{Q}_\varepsilon$,
computer-assisted	the remaining compact set.

Here

$$(44) \quad \mathcal{Q}_\varepsilon := \{(t, d) : 0.352 \leq t \leq 0.450, 0 \leq d \leq 0.005\}.$$

For the first region, Proposition A.4 gives $c_\varepsilon > 0.5308462420554 > 1/2$. From (43),

$$F_\varepsilon(t, \xi) = c_\varepsilon G(2t) + m_{\text{cert}}(1 - \theta)\beta \Gamma_5(u) \Gamma_5(v) + m_{\text{cert}}\theta \Phi(u) \Phi(v) \\ - \frac{G(u) + G(v)}{2}.$$

The two middle terms are nonnegative. Since $(u + v)/2 = t$ and G is concave by Lemma 5.2,

$$\frac{G(u) + G(v)}{2} \leq G(t).$$

Consequently

$$F_\varepsilon(t, \xi) \geq c_\varepsilon G(2t) - G(t).$$

Define

$$\eta_\varepsilon(t) := 2c_\varepsilon(1 - \log(2t)) - 1 - e^t(-\log t + t).$$

Using (34) and (35) gives

$$(45) \quad F_\varepsilon(t, \xi) \geq t \eta_\varepsilon(t).$$

Differentiation gives

$$\eta'_\varepsilon(t) = \frac{e^t - 2c_\varepsilon}{t} - e^t(1 - \log t + t).$$

For $0 < t \leq 10^{-5}$, $e^t < 1.00002 < 2c_\varepsilon$ and $1 - \log t + t > 0$, so both terms are negative. Hence $\eta'_\varepsilon(t) < 0$, and Proposition A.4 gives

$$\eta_\varepsilon(10^{-5}) \geq 0.035919173514281852 > 0.$$

Thus (45) is positive for $0 < t \leq 10^{-5}$.

For $t \geq 8$, discard the two nonnegative terms as above. Lemma 5.2 gives

$$G(2t) \geq 2t + 1 - e^{-2t}, \quad G(u) \leq u + 1, \quad G(v) \leq v + 1.$$

Since $u + v = 2t$,

$$\frac{G(u) + G(v)}{2} \leq t + 1,$$

and therefore

$$F_\varepsilon(t, \xi) \geq c_\varepsilon(2t + 1 - e^{-2t}) - (t + 1).$$

The derivative of $c_\varepsilon(2t + 1 - e^{-2t}) - (t + 1)$ is

$$2c_\varepsilon + 2c_\varepsilon e^{-2t} - 1 > 2c_\varepsilon - 1 > 0.$$

At $t = 8$ this lower bound is

$$c_\varepsilon(17 - e^{-16}) - 9,$$

which is at least 0.024386055203189365 by Proposition A.4. Hence $F_\varepsilon > 0$ for every $t \geq 8$.

For the third analytic region, again discard the two nonnegative terms. Because $u \leq 2t$ and G is increasing,

$$\frac{1}{2}G(u) \leq \frac{1}{2}G(2t).$$

It follows that

$$F_\varepsilon(t, \xi) \geq \left(c_\varepsilon - \frac{1}{2}\right)G(2t) - \frac{1}{2}G(v).$$

Thus every point satisfying $(c_\varepsilon - \frac{1}{2})G(2t) - \frac{1}{2}G(v) > 0$ is settled analytically.

In particular, Proposition A.4 gives positivity of this lower bound at $t = 10^{-5}$ and $v = 10^{-7}$. Since $c_\varepsilon > 1/2$ and G is increasing, the lower bound is increasing in t and decreasing in v . Hence the region $t \geq 10^{-5}$, $0 \leq v \leq 10^{-7}$ is contained in this third analytic region.

It remains to cover the compact points outside these three analytic regions. On $\mathcal{Q}_\varepsilon$, Proposition A.5 gives

$$\mathcal{N}_{tt} \geq 0.22836139797078275, \quad \mathcal{N}_{dd} \geq 0.68037764661274025,$$

and

$$\mathcal{N}_{tt}\mathcal{N}_{dd} - \mathcal{N}_{td}^2 \geq 0.17291110546253904 > 0.$$

Hence the Hessian is positive definite and $\mathcal{N}$ is strictly convex on $\mathcal{Q}_\varepsilon$. The function is even in d , so its d -derivative vanishes on $d = 0$. The same proposition gives

$$\partial_t \mathcal{N}(0.35615880, 0) < 0, \quad \partial_t \mathcal{N}(0.35615895, 0) > 0.$$

By continuity there is a zero of the t -derivative between these values, and strict convexity makes the resulting stationary point unique in $\mathcal{Q}_\varepsilon$. Proposition A.5 also gives

$$\mathcal{N}(t, 0) \geq 6.7220206068085965 \cdot 10^{-7} > 0 \quad (0.35615880 \leq t \leq 0.35615895).$$

Thus the minimum on $\mathcal{Q}_\varepsilon$ is positive.

For the remaining compact set, use the boxes and the centered mean-value estimate defined before (39), now for F_ε . Although F_ε is only piecewise C^1 , the same mean-value estimate remains valid by splitting each coordinate segment at the finitely many breakpoints and bounding the derivative on each resulting piece. Proposition A.6 gives a finite cover on which

$$F_\varepsilon(t_0, \xi_0) - \sup |\partial_t F_\varepsilon| \Delta_t - \sup |\partial_\xi F_\varepsilon| \Delta_\xi > 0.$$

The smallest certified lower bound is $5.3640462249763984 \cdot 10^{-9}$. The five regions in the table cover all $t > 0$ and $0 \leq \xi \leq 1$, so $F_\varepsilon(t, \xi) \geq 0$ everywhere.

Returning to $d = t\xi$ proves (42) for $0 < x \leq y < 1$. Symmetry gives the inequality throughout the open square, and continuity gives the boundary cases. $\square$

Remark. The previous proof makes explicit the role of Example 4. Lemma 5.6 gives $F(t_*, 0) = 0$. The interval for x_* in Proposition A.1 lies in $(1/2, x_{\text{cut}}) \cap (1/16, 15/16)$, so the definitions of D , χ , and φ give $\varphi(x_*) > 0$. Since $e^{-t_*} = x_*$, the definition of Φ yields $\Phi(t_*) = e^{t_*}\varphi(x_*) > 0$. Thus Example 4 supplies a positive term at the unique equality configuration of the two-construction estimate. This is a quantitative analogue of the qualitative mechanism in Della Fiore–Costa–Dalai [5, Section 5].

Proof of Theorem 1.2. We verify the hypothesis of Liu’s criterion, Proposition 2.1. Set

$$\varepsilon_0 := \frac{1}{50000} = 2 \cdot 10^{-5}, \quad m_0 := m_* - \varepsilon_0, \quad c := c_L + \varepsilon_0.$$

Let μ be a probability measure on $[0, 1]$ satisfying

$$\int s \mu(ds) \leq c.$$

If $S \sim \mu$, let P be the distribution of $1 - S$. By (4) and $c_L = 1 - m_*$,

$$m(P) = 1 - \int s \mu(ds) \geq 1 - c = m_0.$$

Moreover $H(1-s) = H(s)$ gives

$$\int H \, dP = \int H \, d\mu.$$

Put

$$\gamma := \frac{m_0}{m_{\text{cert}}}.$$

Since

$$m_0 - m_{\text{cert}} = \varepsilon_{\text{cert}} - \varepsilon_0 = 5 \cdot 10^{-7} > 0,$$

we have $\gamma > 1$.

Apply Proposition 2.1 with the three protocols $\Pi^{(0)}$, $\Pi^{(5)}$, and $\Pi^{(4)}$, with weights

$$w_0 := (1-\theta)\alpha, \quad w_5 := (1-\theta)\beta, \quad w_4 := \theta.$$

They are nonnegative and

$$w_0 + w_5 + w_4 = (1-\theta)(\alpha + \beta) + \theta = 1.$$

For the three admissible classes take

$$\mathcal{K}_0(\mu) := \{\mu \otimes \mu\}, \quad \mathcal{K}_5(\mu) := \mathcal{C}_3(\mu), \quad \mathcal{K}_4(\mu) := \mathcal{C}_3(\mu).$$

The independent protocol induces $\mu \otimes \mu$, while Examples 5 and 4 are conditionally i.i.d., so their induced laws belong to $\mathcal{C}_3(\mu)$.

Integrate Lemma 6.3 with respect to $P(dx)P(dy)$. For the right-hand side, Fubini's theorem gives

$$\begin{aligned} & \iint (xH(y) + yH(x)) P(dx)P(dy) \\ &= \left(\int x P(dx) \right) \left(\int H(y) P(dy) \right) + \left(\int y P(dy) \right) \left(\int H(x) P(dx) \right) \\ &= 2m(P) \int H \, dP. \end{aligned}$$

Dividing the integrated inequality by 2 therefore yields

$$(46) \quad \begin{aligned} & (1-\theta) \left(\alpha \iint H(xy) P(dx)P(dy) + \beta \left(\int b \, dP \right)^2 \right) \\ & + \theta \left(\int \varphi \, dP \right)^2 \geq \frac{m(P)}{m_{\text{cert}}} \int H \, dP. \end{aligned}$$

Under the reflection $x = 1-s$, $y = 1-t$, the term of $\Pi^{(0)}$ in Liu's criterion becomes

$$\iint H(xy) P(dx)P(dy).$$

For $\Pi^{(5)}$, reflection sends $\mathcal{C}_3(\mu)$ bijectively to $\mathcal{C}_3(P)$, and Lemma 3.3 gives

$$\inf_{\rho \in \mathcal{C}_3(P)} \iint H(r(x, y)) \rho(dx, dy) \geq \left(\int b \, dP \right)^2.$$

For $\Pi^{(4)}$, the same reflection and Proposition 6.2 give

$$\inf_{\rho \in \mathcal{C}_3(P)} \iint K_4(x, y) \rho(dx, dy) \geq \left(\int \varphi \, dP \right)^2.$$

Multiplying these three estimates by w_0, w_5, w_4 and using (46), the left-hand side of Liu's criterion is at least

$$\frac{m(P)}{m_{\text{cert}}} \int H \, dP \geq \gamma \int H \, dP = \gamma \int H \, d\mu.$$

Here the second inequality uses $m(P) \geq m_0$ and $\int H \, dP \geq 0$. Thus all hypotheses of Proposition 2.1 hold, and it gives $c \leq c_{\text{UC}}$. Therefore

$$c_{\text{UC}} \geq c_{\text{L}} + 2 \cdot 10^{-5} > 0.382729087918735,$$

which proves Theorem 1.2. $\square$

APPENDIX A. COMPUTATIONAL RESULTS

The goal of this appendix is to state, in mathematical form, the finite computations used in the proof. Propositions A.1 and A.3–A.6 are verified with outward-rounded interval arithmetic, while Proposition A.2 is an exact rational calculation. The computational supplement explains the verification procedures and identifies the ancillary source files.

The first proposition contains the numerical constants used in Section 3 and the scalar bounds, local Hessian bounds, and finite cover used in Section 5. Its verification is documented in Section 1 of the computational supplement.

Proposition A.1. The constants of Section 3 satisfy

$$\begin{aligned} x_* &\in [0.69078759392498, 0.69078759392500], \\ m_* &\in [0.61729091208117, 0.61729091208136], \\ \beta &\in [0.1000525598619, 0.1000525598639], \\ t_* &\in [0.36992289186769, 0.36992289186773]. \end{aligned}$$

The same certificate directly verifies $c_L > 0.382709087918735$. Here $c_0 = m_*\alpha$ and

$$\eta(t) = 2c_0(1 - \log(2t)) - 1 - e^t(-\log t + t),$$

as in the proof of Lemma 5.3. The scalar lower bounds used in Lemmas 5.3 and 5.4 are

$$\begin{aligned} \eta(0.002) &\geq 0.016672474402352748, \\ \left(c_0 - \frac{1}{2}\right) G(0.004) - \frac{1}{2} G(5 \cdot 10^{-5}) &\geq 0.0011788376979667077, \\ \left(c_0 - \frac{1}{2}\right) G(2.8) - \frac{1}{2} G(0.1) &\geq 0.035598095197518845, \\ c_0(9.2 - e^{-8.2}) - 5.1 &\geq 0.010717682423498683. \end{aligned}$$

On the rectangle (38),

$$\begin{aligned} F_{tt} &\geq 0.13459156779665421, \\ F_{dd} &\geq 0.87181314699157286, \\ F_{tt}F_{dd} - F_{td}^2 &\geq 0.050114365218749776. \end{aligned}$$

For $a < b$ and $\nu > 0$, define

$$\mathcal{R}(a, b; \nu) := \{(t, \xi) : a \leq t \leq b, 0 \leq \xi \leq 1, t(1 - \xi) \geq \nu\}.$$

Each of the five regions

$$\begin{aligned} \mathcal{R}(0.002, 0.1; 5 \cdot 10^{-5}), & \quad \mathcal{R}(0.1, 0.34; 5 \cdot 10^{-5}), \\ \mathcal{R}(0.34, 0.40; 5 \cdot 10^{-5}), & \quad \mathcal{R}(0.40, 1.4; 5 \cdot 10^{-5}), \\ \mathcal{R}(1.4, 4.1; 0.1) & \end{aligned}$$

has a finite cover by boxes of the form defined before (39). After the substitution $d = t\xi$, every box either maps into the local rectangle $\mathcal{Q}_0$ defined in (38), or has a positive lower bound from (39). The smallest lower bounds in the five regions, in the listed order, are

$$\begin{aligned} 1.4683343626197839 \cdot 10^{-4}, \quad 1.2396204165999228 \cdot 10^{-6}, \quad 5.6012230503014744 \cdot 10^{-4}, \\ 6.9620924720995344 \cdot 10^{-6}, \quad 4.4688168746738737 \cdot 10^{-3}. \end{aligned}$$

The following proposition records the exact Bernstein coefficients used in Lemma 4.3. Its verification is documented in Section 2 of the computational supplement.

Proposition A.2. After the rescaling $x = (43/64)s$, $y = (43/64)t$, write

$$B_i^3(s) := \binom{3}{i} s^i (1-s)^{3-i}, \quad B_j^3(t) := \binom{3}{j} t^j (1-t)^{3-j}.$$

The polynomial $WUV - (1-r)$ has bidegree $(3, 3)$ and, in the product Bernstein basis $\{B_i^3(s)B_j^3(t) : 0 \leq i, j \leq 3\}$, its coefficient matrix is

$$\begin{pmatrix} 7 & \frac{41}{12} & \frac{8221}{6144} & \frac{20653}{131072} \\ \frac{41}{12} & \frac{18803}{9216} & \frac{402175}{393216} & \frac{1894129}{6291456} \\ \frac{8221}{6144} & \frac{402175}{393216} & \frac{52130497}{75497472} & \frac{587913113}{1610612736} \\ \frac{20653}{131072} & \frac{1894129}{6291456} & \frac{587913113}{1610612736} & \frac{6389233321}{17179869184} \end{pmatrix}.$$

Every coefficient is strictly positive.

The next proposition is the finite lower bound used for Liu's Example 4. Its verification is documented in Section 3 of the computational supplement.

Proposition A.3. The triangle

$$\{(x, y) : 1/64 \leq x \leq y \leq 63/64\}$$

has a finite subdivision consistent with the intervals in the following table:

function	interval	formula
$\tilde{a}(x)$	$0 \leq x \leq 1/2$	1
$\tilde{a}(x)$	$1/2 < x < 1/\sqrt{2}$	$\sqrt{\frac{1-2x^2}{2x(1-x)}}$
$\tilde{a}(x)$	$1/\sqrt{2} \leq x \leq 1$	0
$\chi(x)$	$0 \leq x \leq 1/64$	0
$\chi(x)$	$1/64 < x < 1/16$	$\frac{x-1/64}{3/64}$
$\chi(x)$	$1/16 \leq x \leq 15/16$	1
$\chi(x)$	$15/16 < x < 63/64$	$\frac{63/64-x}{3/64}$
$\chi(x)$	$63/64 \leq x \leq 1$	0
$D(x)$	$0 \leq x \leq 1/2$	$H(x)$
$D(x)$	$1/2 \leq x \leq x_{\text{cut}}$	$\log 2$
$D(x)$	$x_{\text{cut}} \leq x \leq 1$	$\frac{\log 2}{H(x_{\text{cut}}^2)} H(x^2)$

At the irrational breakpoint $1/\sqrt{2}$, the subdivision uses the binary64 approximation obtained by evaluating $\sqrt{1/2}$ with MPFR and rounding upward. The cell meeting this breakpoint is certified by bounds valid on both sides of the exact value, so this approximation leaves no uncovered points; see Section 3 of the computational supplement. On every resulting cell, direct outward-rounded interval evaluation or, when necessary, the two-variable mean-value estimate gives

$$K_4(x, y) - \varphi(x)\varphi(y) \geq 2.3311935604120031 \cdot 10^{-8}.$$

The function on the left is symmetric in x, y , so the same inequality holds on the reflected triangle $y \leq x$. Outside $[1/64, 63/64]^2$, at least one cutoff factor is zero, hence $\varphi(x)\varphi(y) = 0$, while $K_4(x, y) \geq 0$. Therefore (40) holds on $[0, 1]^2$.

Section 4 of the computational supplement documents the scalar evaluations used below.

Proposition A.4. With the parameters in (41), define

$$c_\varepsilon := m_{\text{cert}}(1 - \theta)\alpha, \quad \eta_\varepsilon(t) := 2c_\varepsilon(1 - \log(2t)) - 1 - e^t(-\log t + t).$$

Then $c_\varepsilon > 0.5308462420554$. The scalar checks used in the three analytic regions of the proof of Lemma 6.3 are

$$\begin{aligned} \eta_\varepsilon(10^{-5}) &\geq 0.035919173514281852, \\ c_\varepsilon(17 - e^{-16}) - 9 &\geq 0.024386055203189365, \\ \left(c_\varepsilon - \frac{1}{2}\right) G(2 \cdot 10^{-5}) - \frac{1}{2} G(10^{-7}) &\geq 6.4360829338080744 \cdot 10^{-6}. \end{aligned}$$

Section 5 of the computational supplement documents the local Hessian and derivative evaluations for $\mathcal{N}$.

Proposition A.5. On the rectangle (44), the Hessian of $\mathcal{N}$ satisfies

$$\begin{aligned} \mathcal{N}_{tt} &\geq 0.22836139797078275, \\ \mathcal{N}_{dd} &\geq 0.68037764661274025, \\ \mathcal{N}_{tt}\mathcal{N}_{dd} - \mathcal{N}_{td}^2 &\geq 0.17291110546253904. \end{aligned}$$

On $d = 0$,

$$\begin{aligned} \partial_t \mathcal{N}(0.35615880, 0) &\leq -1.6551140458176399 \cdot 10^{-8}, \\ \partial_t \mathcal{N}(0.35615895, 0) &\geq 2.8118104733465537 \cdot 10^{-8}. \end{aligned}$$

For every $t \in [0.35615880, 0.35615895]$,

$$\mathcal{N}(t, 0) \geq 6.7220206068085965 \cdot 10^{-7}.$$

Section 6 of the computational supplement documents the finite complementary cover.

Proposition A.6. Let

$$F_\varepsilon(t, \xi) := \mathcal{N}(t, t\xi), \quad t > 0, \quad 0 \leq \xi \leq 1.$$

Remove from this domain the small- t region $0 < t \leq 10^{-5}$, the large- t region $t \geq 8$, every point satisfying

$$\left(c_\varepsilon - \frac{1}{2}\right) G(2t) - \frac{1}{2} G(t(1 - \xi)) > 0,$$

and the local rectangle (44) after $d = t\xi$. The remaining set has a finite cover by boxes. On every box, the centered mean-value lower bound described in (39), with F_ε in place of $\tilde{F}$, is positive. The smallest such lower bound is

$$5.3640462249763984 \cdot 10^{-9}.$$

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